

03.19.2026



/ Agenda

Introductions

Meeting goals/objectives

Import SDK

- MCUXPresso Software download
- MCUXpresso IDE
- How to program code

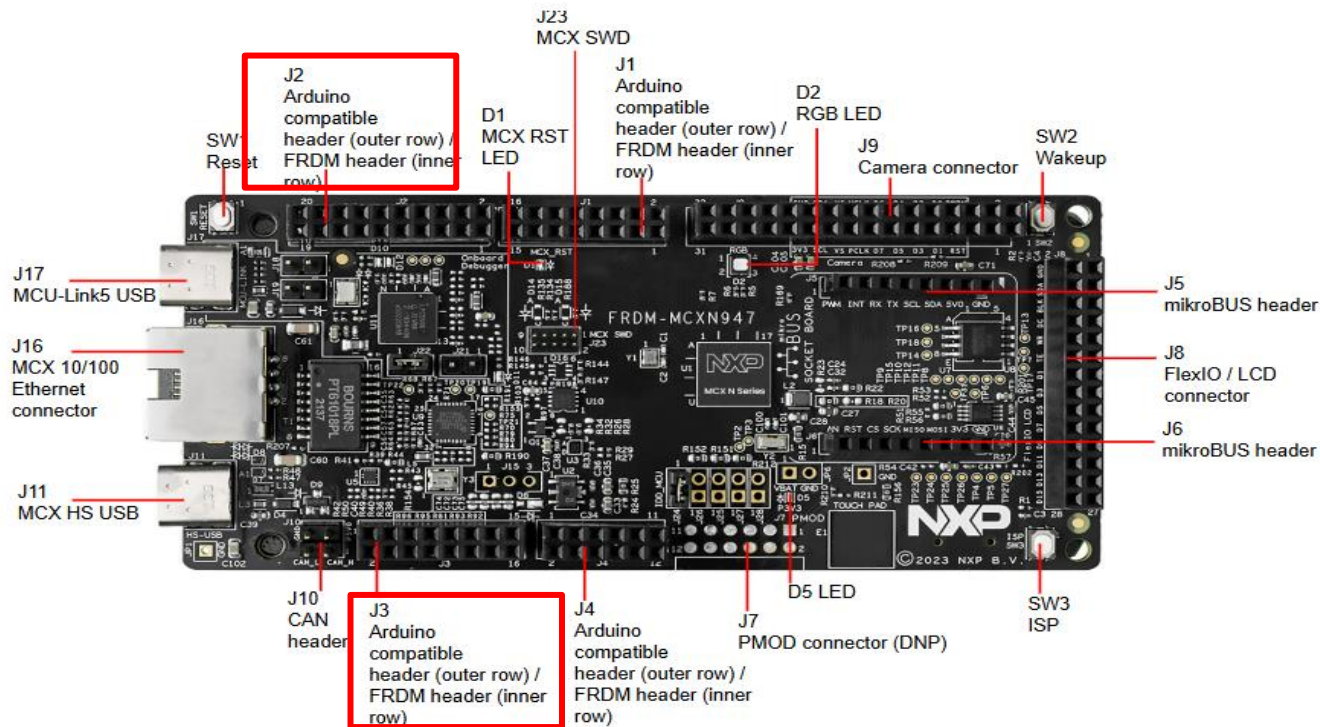
Get To Know MCXN947

NXP Wi-Fi® Modules Using MCXN947 Platform

Q&A

Get To Know MCXN947

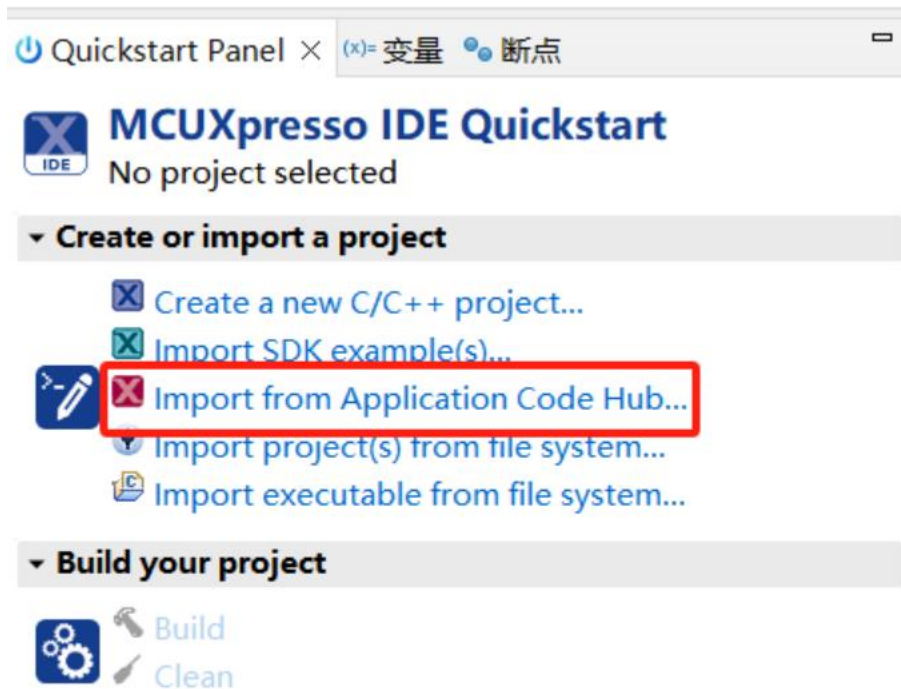
Get Started with the FRDM-MCXN947



FRDM-MCXN947 connectors, push buttons, and LEDs (top-side view)

Import SDK Example

/ Import SDK Example --- Open MCUXpresso IDE



Quick Start Panel, choose Import from Application Code Hub

/ Import SDK

Import Projects from Application Code Hub

Browse, clone and import application code hub projects. Select a project and click "Copy GitHub link" to continue.

The screenshot displays the NXP Application Code Hub interface. On the left, a sidebar titled "Device Families" lists various NXP microcontroller families: DSC, i.MX, i.MX RT, K32W, Kinetis, KW45, LPC, MCX (selected with a green checkmark), and MW. Below this list is a "Clear All Properties" button. The main content area features a search bar with the text "Wifi" and a red rectangular box highlighting it. Below the search bar, there are filters for "Device Families" (MCX) and "Toolchains" (MCUXpresso IDE). A "Search" button with the text "Wifi" is also present. The search results are sorted by "Recently Added" and show a total of 3 items. The first result is titled "Wi-Fi CLI M2 FRDM MCXN947" and is for the FRDM-MCXN947 board. It includes a description: "This is an example of cli use with wifi on frdm-mcxn947". A red rectangular box highlights the "Wireless Connectivity" tag associated with this project. The second result is titled "Wi-Fi connect using LCD interface on FRDM-MCXN947 using Wi-Fi expansion board FRDM-IW416-AW-AM510" and is for the FRDM-MCXN947 board. It includes a description: "This is a demo example of Wi-Fi connect using LCD interface on FRDM-MCXN947 using Wi-Fi".

NXP Application Code Hub

Expansion Board Hub

Device Families

- ☐ DSC
- ☐ i.MX
- ☐ i.MX RT
- ☐ K32W
- ☐ Kinetis
- ☐ KW45
- ☐ LPC
- ☒ MCX
- ☐ MW

Clear All Properties

Search: Wifi

Sort: Recently Added

Content

Total: 3

Device Families: MCX

Toolchains: MCUXpresso IDE

Search: "Wifi"

Wi-Fi CLI M2 FRDM MCXN947
FRDM-MCXN947
This is an example of cli use with wifi on frdm-mcxn947
Wireless Connectivity RTOS

Wi-Fi connect using LCD interface on FRDM-MCXN947 using Wi-Fi expansion board FRDM-IW416-AW-AM510
FRDM-MCXN947
This is a demo example of Wi-Fi connect using LCD interface on FRDM-MCXN947 using Wi-Fi

/ Import SDK example

[Visit on GitHub](#)[GitHub link](#)[Share link](#)

Boards



NXP Application Code Hub



WiFi CLI Azure Wave AW-AM510 FRDM MCXN947

This is an example of cli use with wifi running on serial terminal in frdm-mcxn947, this example has many test options for wifi. There are the options:

```
-help  
-wlan-version
```


/ Import SDK example

Local Destination

Configure the local storage location for dm-wifi-cli-frdm-mcxn947.



Destination

Directory:

Initial branch:

☐ Clone submodules

Configuration

Remote name:

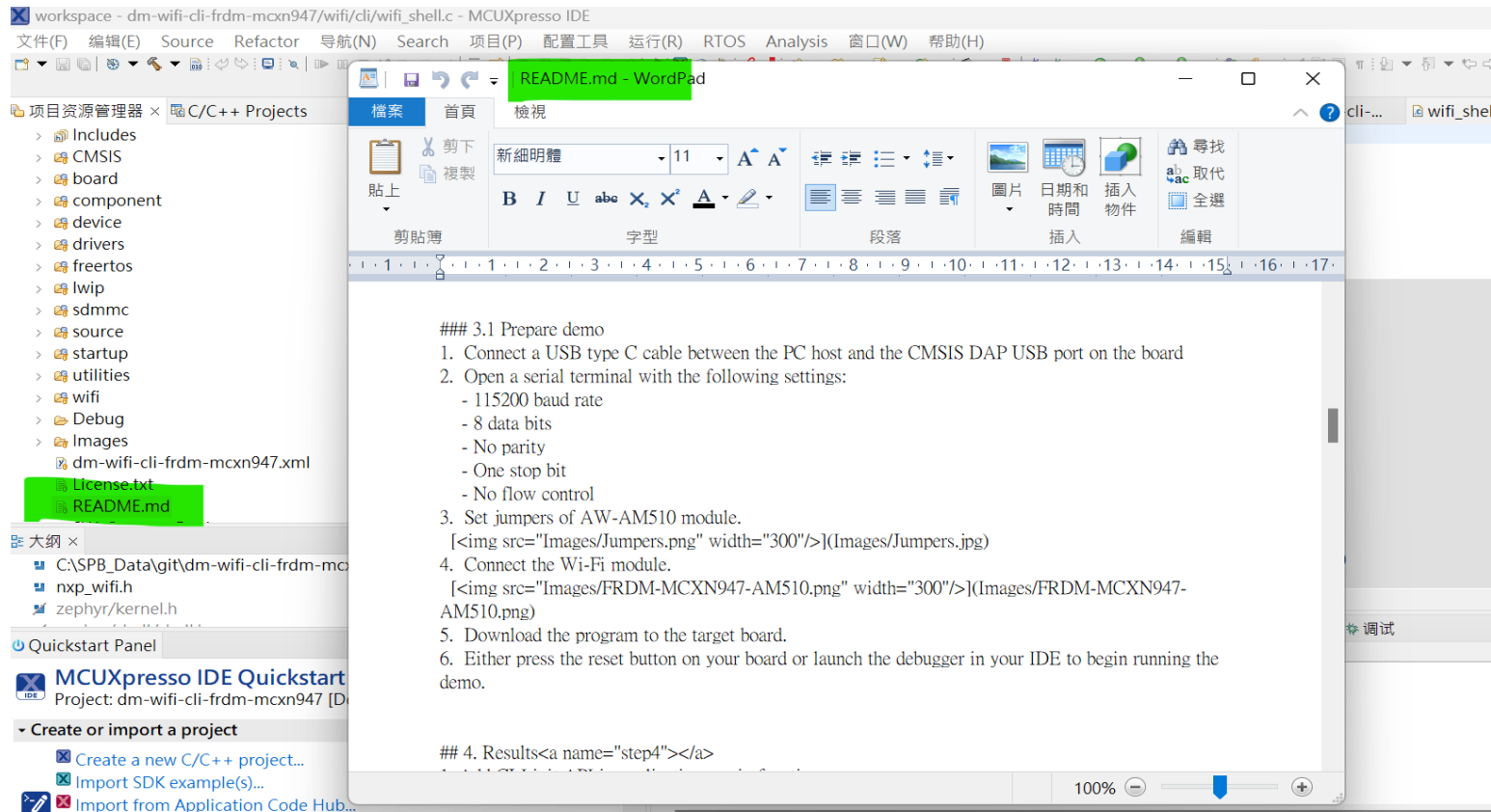
Import SDK example

The screenshot displays the MCUXpresso IDE interface with the following components:

- Project Explorer:** A red box highlights the project `dm-wifi-cli-frdm-mcxn947` under the `C/C++ Projects` folder. The project is currently in `<Debug>` mode.
- Source Explorer:** Shows the project's file structure, including `includes`, `cmsis`, `board`, `component`, `device`, `drivers`, `freertos`, `lwip`, `sdmmc`, `source`, `startup`, `utilities`, `wifi`, `certs`, and `cli`.
- Code Editor:** Displays the `wifi_shell.c` file. The code includes headers like `<zephyr/kernel.h>` and `<zephyr/shell/shell.h>`, and defines a `cli_command` structure.
- Quickstart Panel:** Located at the bottom left, it provides instructions for creating or importing a project. The `Import SDK example(s)...` option is selected.
- Console:** At the bottom right, it shows the output of the flash tool, indicating successful flashing of the `dm-wifi-cli-frdm-mcxn947` project.

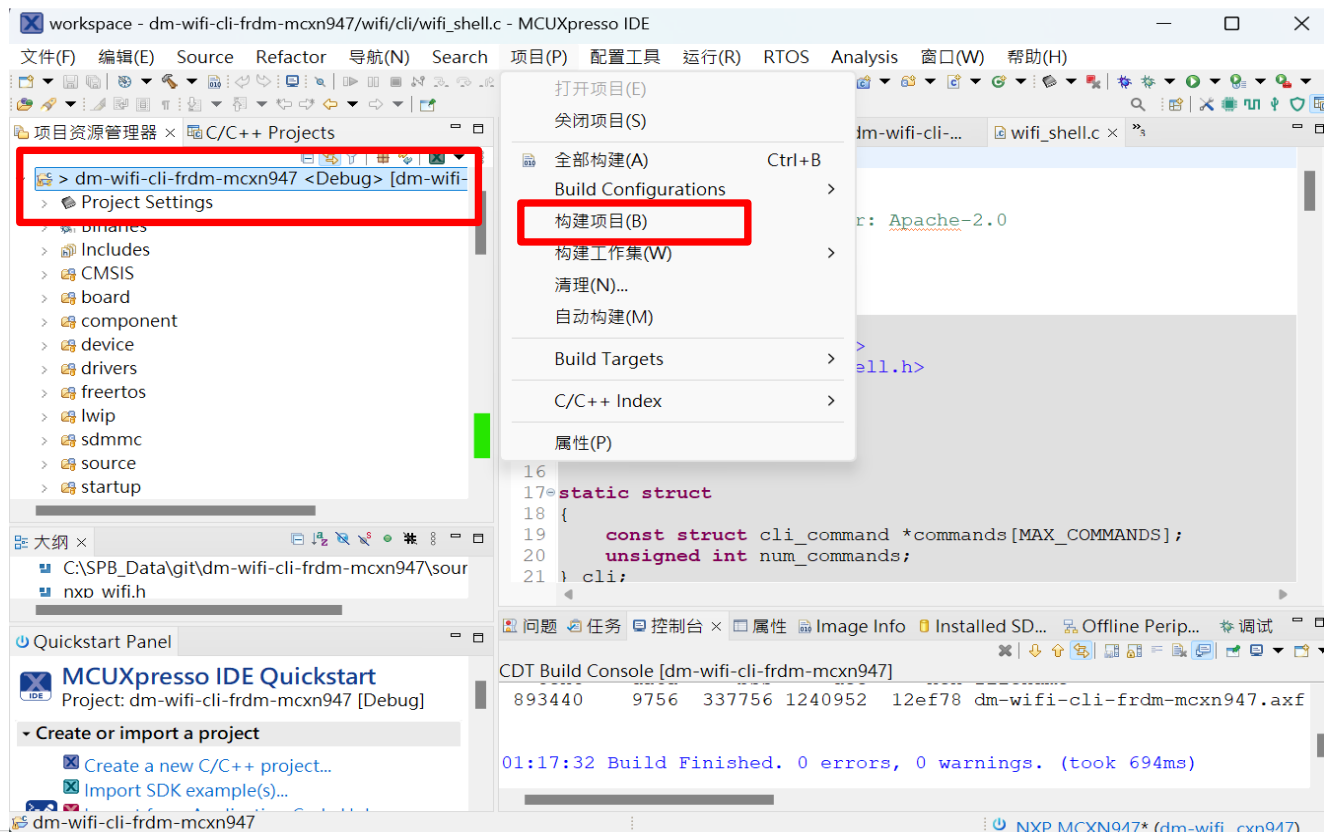
Environment setting and Programming

Build and Run Wi-Fi demo from the SDK



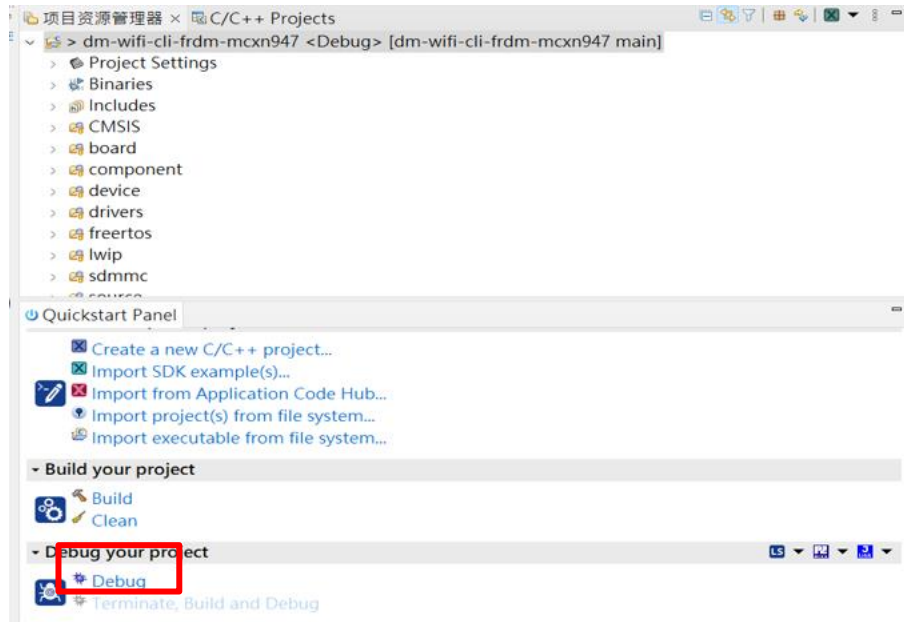
/ How to build

選擇專案右鍵點擊 Build Project, 或是直接點擊

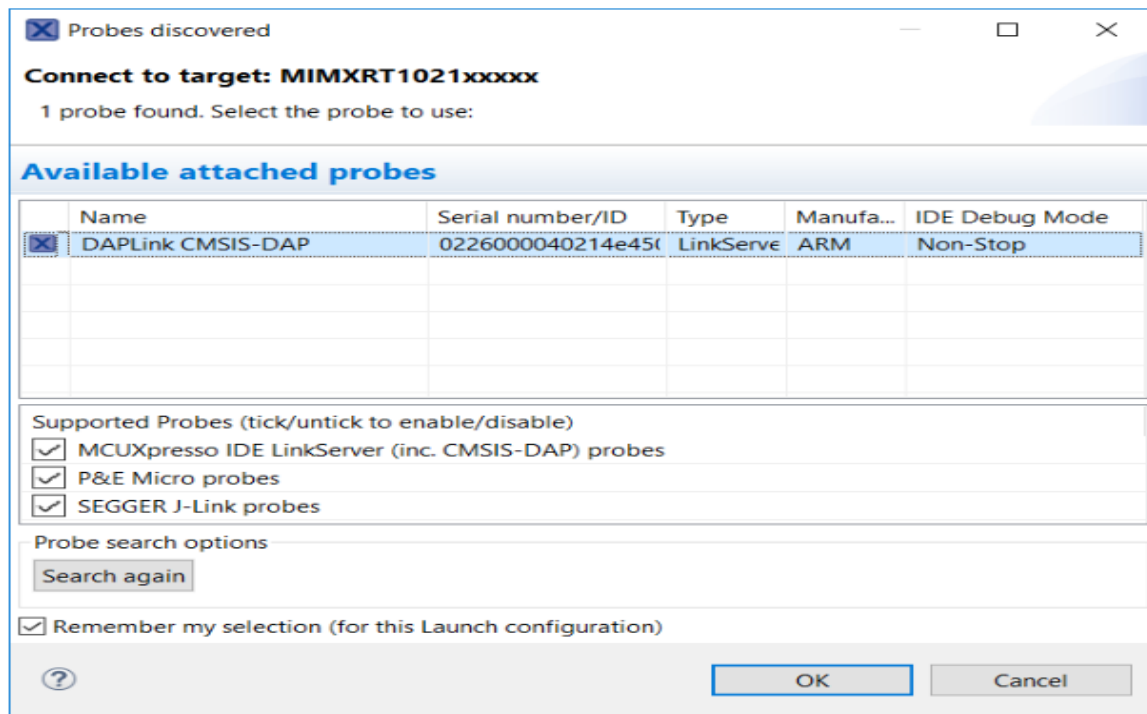


/ Programming code

1. Now that the project has been compiled , you can flash and run it
2. Make sure USB cable plugged into the debug connect on the MCXN947 – EVK , and click on Debug in the QuickStart Panel .



/ Programming code



Getting Started with NXP Wi-Fi® Modules Using MCXN Platform

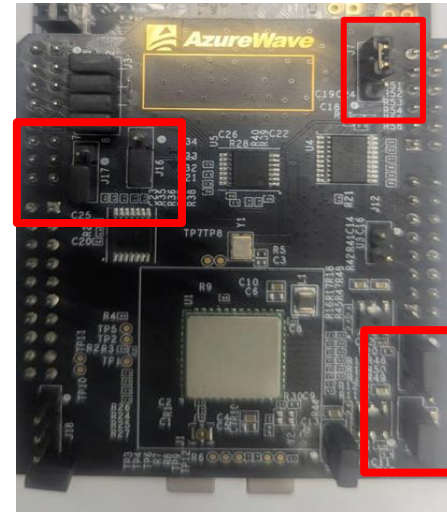
/FRDM-IW416-AW-AM510

Wireless Connectivity

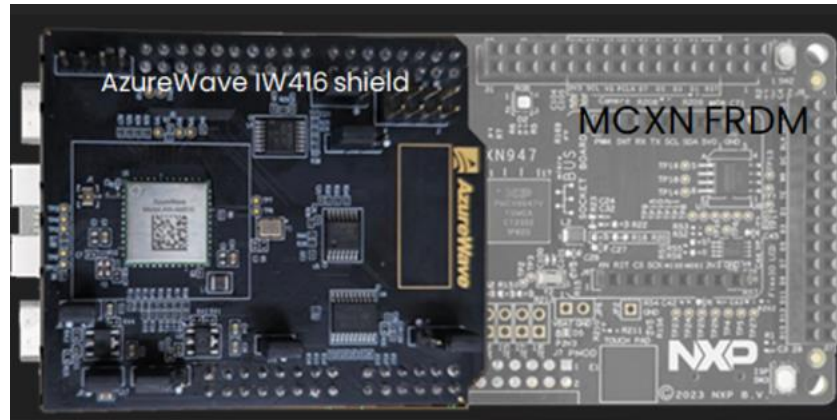
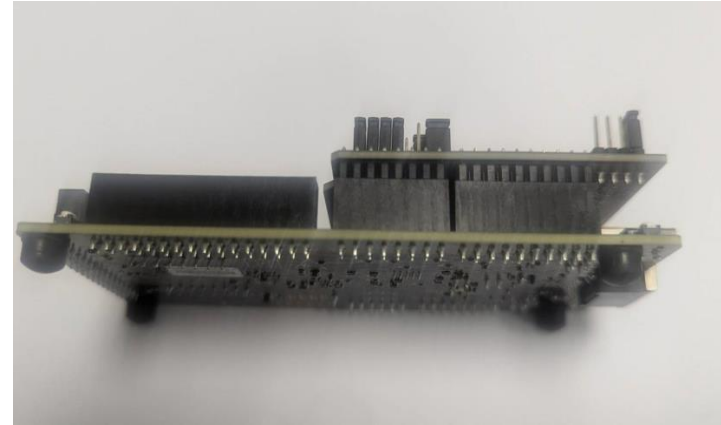
Wi-Fi® + Bluetooth® + 802.15.4

- **IW416**: 2.4/5 GHz Dual-Band 1x1 Wi-Fi® 4 (802.11n) + Bluetooth® 5.2 Solution

Set jumpers of AW-AM510 module.



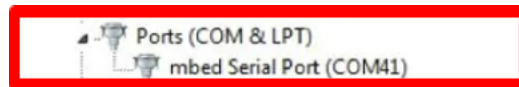
Wi-Fi® Modules Using FRDM-MCXXN947 Platform



1. **Download** Tera Term from SourceForge. After the download, run the installer and then return to this webpage to continue.

[DOWNLOAD](#)

2. **Launch** Tera Term. The first time it launches, it will show you the following dialog. Select the serial option. Assuming your board is plugged in, there should be a COM port automatically populated in the list.



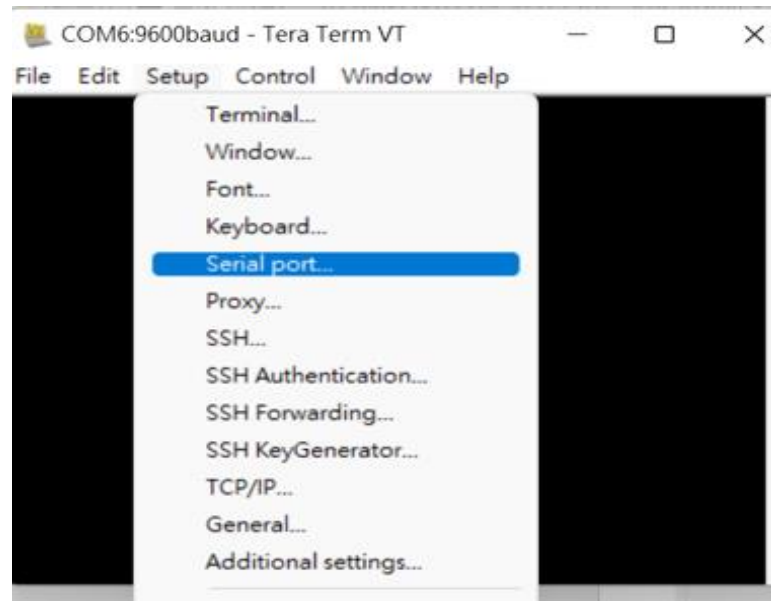
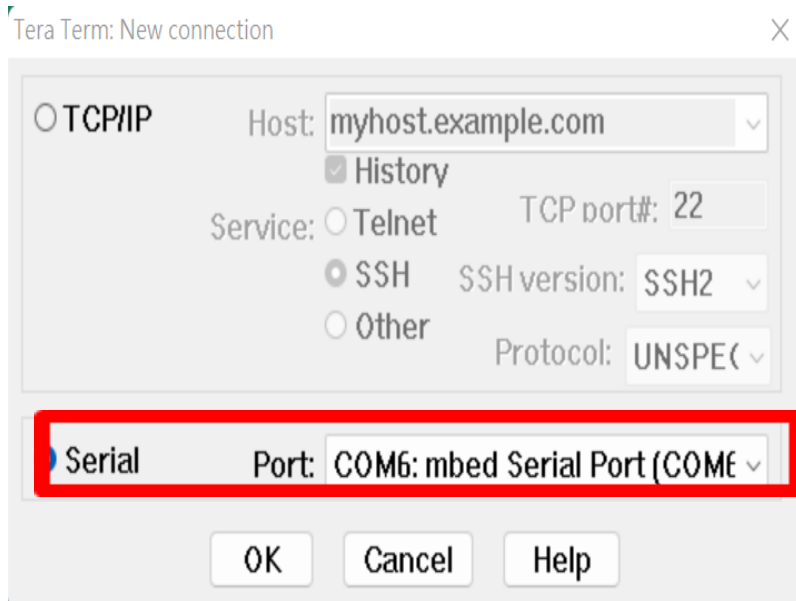
Not sure how to use a terminal application? Try one of these tutorials:

[Tera Term Tutorial](#)

[PuTTY Tutorial](#)

3. **Configure** the serial port settings (using the COM port number identified earlier) to 115200 baud rate, 8 data bits, no parity and 1 stop bit. To do this, go to Setup -> Serial Port and change the settings.

• **Download** [Tera Term](#). After the download, run the installer and then return to this webpage to continue.



Tera Term: Serial port setup

Port: COM6

Baud rate: 115200

Data: 8 bit

Parity: none

Stop: 1 bit

Flow control: none

Transmit delay

0 msec/char 0 msec/line

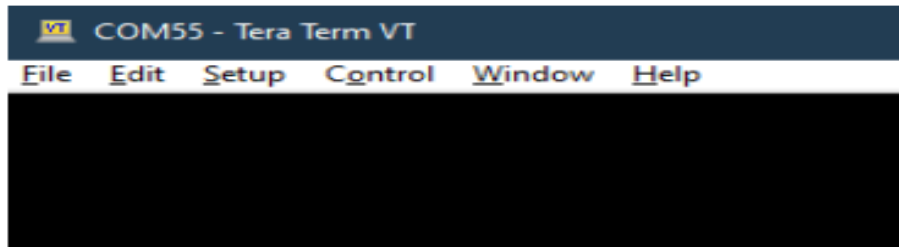
OK

Cancel

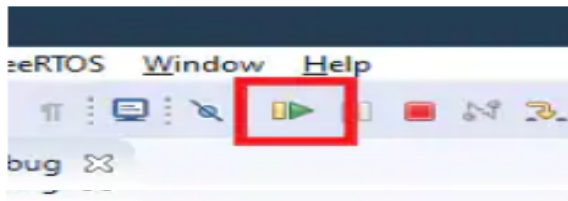
Help

/ Building and running a Wi-Fi demo

Before you start your application, open a Serial Terminal application (ie. PuTTY, TeraTerm). You can find instructions to set up a Terminal application on the EVK's Getting Started website → Section 2: Get Software → PC Configuration

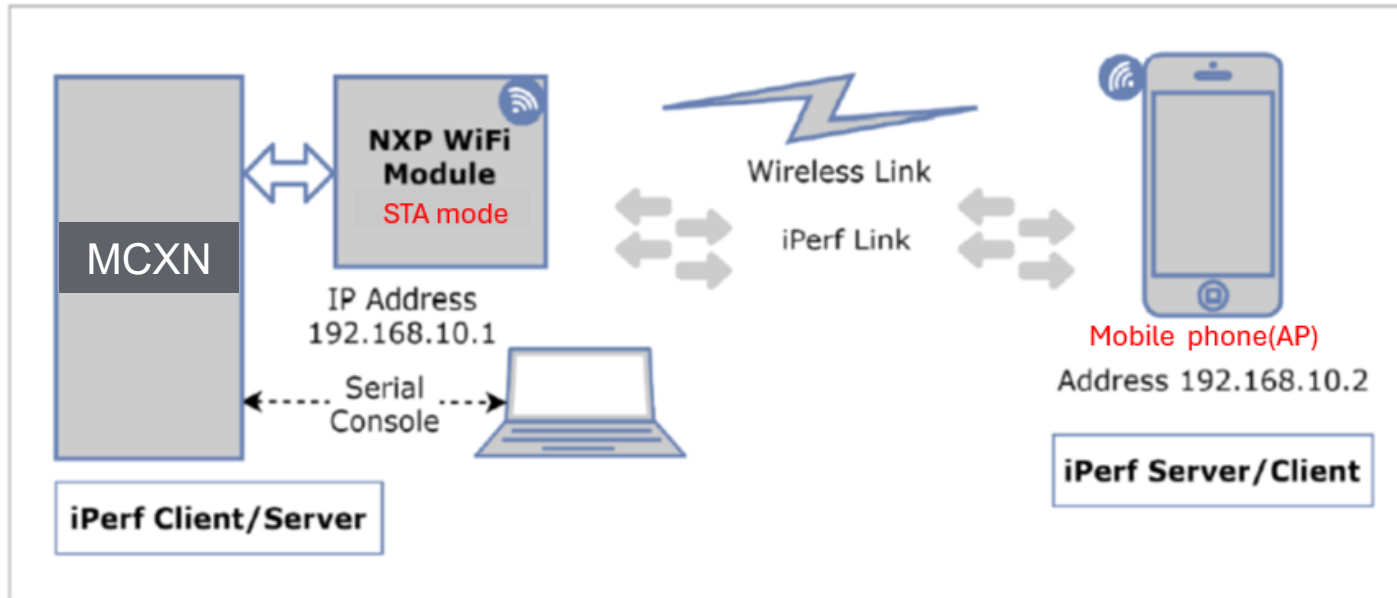


5. Click on the Resume button to start the application

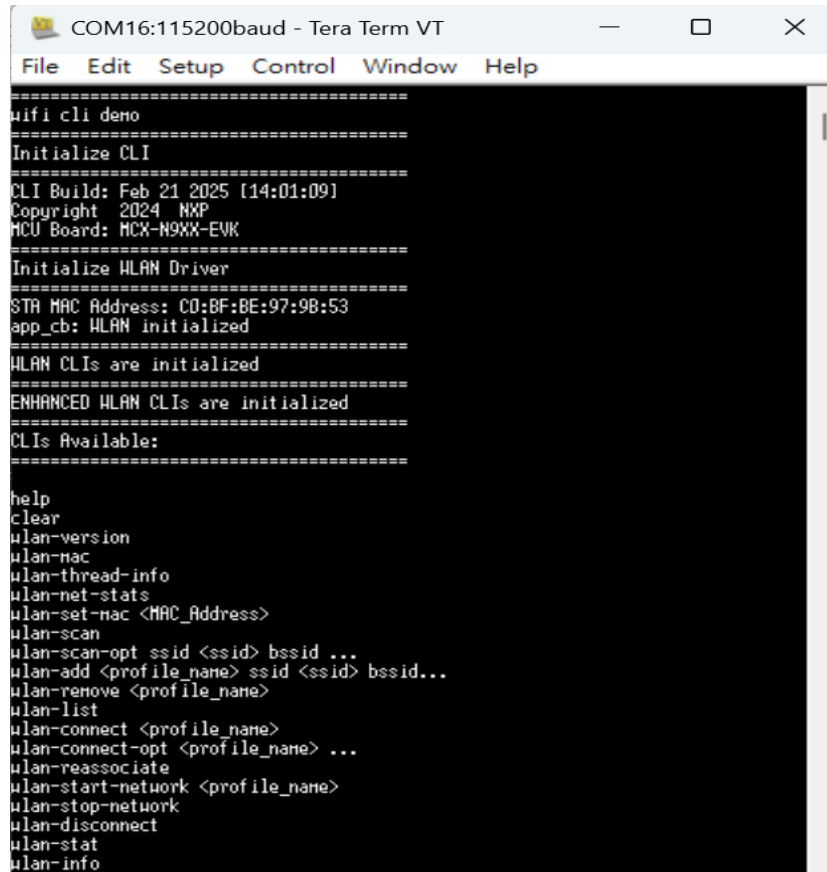


/ STA/uAP mode

Wi-Fi AP



Running a Wi-Fi demo



The screenshot shows a terminal window titled "COM16:115200baud - Tera Term VT". The window contains the following text:

```
=====
wifi cli demo
=====
Initialize CLI
=====
CLI Build: Feb 21 2025 [14:01:09]
Copyright 2024 NXP
MCU Board: MCX-N9XX-EVK
=====
Initialize WLAN Driver
=====
STA MAC Address: C0:BF:BE:97:9B:53
app_cb: WLAN initialized
=====
WLAN CLIs are initialized
=====
ENHANCED WLAN CLIs are initialized
=====
CLIs Available:
=====
help
clear
wlan-version
wlan-mac
wlan-thread-info
wlan-net-stats
wlan-set-mac <MAC_Address>
wlan-scan
wlan-scan-opt ssid <ssid> bssid ...
wlan-add <profile_name> ssid <ssid> bssid...
wlan-remove <profile_name>
wlan-list
wlan-connect <profile_name>
wlan-connect-opt <profile_name> ...
wlan-reassociate
wlan-start-network <profile_name>
wlan-stop-network
wlan-disconnect
wlan-stat
wlan-info
```


/ Reset Wi-Fi module

Command : wlan-version

```
#
# wlan-version
WLAN Driver Version : v1.3.r48.p31
WLAN Firmware Version : IW416-V0, RF878X, FP91, 16.92.21.p142.5, WPA2_CVE_FIX 1,
PVE_FIX 1

# wlan-list
0 networks

# wlan-info
Station not connected
uAP not started

# wlan-address
not connected
```

/ Scan command # wlan-scan

Command : wlan-scan

```
# wlan-scan
wlan-scan wlan-scan-opt wlan-scan-channel-gap
# wlan-scan
Scan scheduled...

# 2 networks found:
C8:78:7D:68:67:93 "R12-6793" Infra
  mode: 802.11N
  channel: 40
  rssi: -81 dBm
  security: WPA2
  WMM: YES
  802.11K: YES
  802.11V: YES
  802.11H: No

52:39:97:EF:69:5B "Byronlink" Infra
  mode: 802.11N
  channel: 48
  rssi: -75 dBm
  security: WPA2
  WMM: YES
  802.11V: YES
  802.11H: Capable
```

此為手機開啟熱點名稱

Add network profile - # wlan-add

```
# wlan-add ?
Usage:
For Station interface
For DHCP IP address assignment:
wlan-add <profile_name> ssid <ssid> [upa2 <psk/psk-sha256/ft-psk> <secret>]
If using WPA2 security, set the PMF configuration as mentioned above
wlan-add <profile_name> ssid <ssid> [owe_only] nfpcc 1 nfpr 1
If using OWE only security, always set the PMF configuration.
NOTE: [log <"19 20 21">] is only supported in Micro-AP mode .
wlan-add <profile_name> ssid <ssid> [upa3 sae <secret>] [pue <0/1/2>] nfpcc <1>
If using WPA3 SAE security, always set the PMF configuration.
wlan-add <profile_name> ssid <ssid> [upa2 psk psk-sha256 <secret>] upa3 sae <secret>
If using WPA2/WPA3 Mixed security, set the PMF configuration as mentioned
For static IP address assignment:
wlan-add <profile_name> ssid <ssid>
ip:<ip_addr>,<gateway_ip>,<netmask>
[bssid <bssid>] [channel <channel number>]
[upa2 <psk/psk-sha256/ft-psk> <secret>] [owe_only] [upa3 sae <secret>] [nfpcc
For Micro-AP interface
wlan-add <profile_name> ssid <ssid>
ip:<ip_addr>,<gateway_ip>,<netmask>
role uap [bssid <bssid>]
[channel <channel number>]
[upa2 <psk/psk-sha256> <secret>] [upa3 sae <secret>] [pue <0/1/2>] [tr <0/1>]
[owe_only]
[nfpcc <0/1>] [nfpr <0/1>]
```

Command : wlan-add ?

```
# wlan-add aaa ssid Byronlink upa2 psk 12345678
Added "aaa"
```

Add profile : aaa

Connect to the Station network using the saved network profile:

```
# wlan-conn
wlan-connect wlan-connect-opt
# wlan-connect aaa
command 'wlan-connect' not found

# wlan-conn
wlan-connect wlan-connect-opt
# wlan-connect aaa
Connecting to network...
Use 'wlan-stat' for current connection status.

# app_cb: WLAN: authenticated to network
# app_cb: WLAN: connected to network
Connected to following BSS:
SSID = [Byronlink]
IPv4 Address: [192.168.30.194]
```

Connect to station

/ Connect to the Station status:

```
Use 'wlan-stat' for current connection status.
```

```
# app_cb: WLAN: authenticated to network
```

```
uapp_cb: WLAN: connected to network
```

```
Connected to following BSS:
```

```
BSSID = [Byronlink]
```

```
IPv4 Address: [192.168.30.194]
```

```
# wlan-st
```

```
wlan-start-network wlan-stop-network wlan-stat
```

```
# wlan-stat
```

```
Station connected (IEEE ps)
```

```
uAP stopped
```

Command : wlan-stat

/ Disconnect to the station status:

```
# wlan-disconnect aaa
# app_cb: disconnected
#
#
# wlan-stat
Station disconnected (Deep sleep)
uAP stopped
#
# wlan-renov
# wlan-remove aaa
Removed "aaa"
```

Command : wlan-disconnect

Command : wlan-remove

Connect to the uAP network using the saved network profile:

```
# wlan-stat
Station disconnected (Deep sleep)
uAP stopped

# wlan-add xyz ssid AP ip:192.168.10.1,192.168.10.1,255.255.255.0 role uap upa2
psk 12345678
Added "xyz"
```

Add uAP profile : xyz

```
# wlan-start-network xyz

# wlaapp_cb: WLAN: UAP Started
=====
Soft AP "AP" started successfully
=====
nDHCp Server started successfully
=====
```

Command : wlan-start-network

```
# wlan-sta
wlan-start-network wlan-stat

# wlan-stat
Station disconnected (Deep sleep)
uAP started (Active)
```

Command : wlan-stat

/ Connect the uAP From PC:



```
wlan-info
Station not connected
uAP started as:
"xyz"

SSID: AP
BSSID: C2:BF:BE:97:9C:53
mode: 802.11n
channel: 1
role: uAP
security: WPA2
wifi capability: 11n
user configure: 11n

IPv4 Address
address: STATIC
IP:          192.168.10.1
gateway:     192.168.10.1
netmask:     255.255.255.0
dns1:        192.168.10.1
dns2:        0.0.0.0

IPv6 Addresses
Link-Local   : FE80::C0BF:BEFF:FE97:9C53 (Tentative)

rssi threshold: 0

# app_cb: WLAN: UAP a Client Associated
=====
Client => 14:85:7F:C6:26:22 Associated with Soft AP
=====
app_cb: WLAN: UAP a Client Connected
=====
Client => 14:85:7F:C6:26:22 Connected with Soft AP
=====
```


/ Ping command format in STA mode

Command : ping [-s <packet_size>] [-c <packet_count>] [-W <timeout in sec>] <ipv4/ipv6 address>

```
ping
Incorrect usage
Usage:
  ping [-s <packet_size>] [-c <packet_count>] [-W <timeout in sec>] <ipv4/ipv6 address>
Default values:
  packet_size: 56
  packet_count: 10
  timeout: 2 sec
# []
```

```
# ping -s 56 -c 2 -W 2 10.163.130.1
PING 10.163.130.1 (10.163.130.1) 56(84) bytes of data
From 10.163.130.194 icmp_seq=1 Destination Host Unreachable
From 10.163.130.194 icmp_seq=2 Destination Host Unreachable

--- 10.163.130.194 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss

# ping -s 56 -c 2 -W 2 10.163.130.194
PING 10.163.130.194 (10.163.130.194) 56(84) bytes of data
64 bytes from 10.163.130.194: icmp_req=1 ttl=255 time=0 ms
64 bytes from 10.163.130.194: icmp_req=2 ttl=255 time=0 ms

--- 10.163.130.194 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss
```

Questions

Thank You

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